GA-02 · Chemistry (NCERT Class 9–12 Level)

RRB JE CBT-2 · Electronics & Allied Engineering Group (CEN 05/2025)

Section: Physics & Chemistry — 15 Questions Combined

> **How to use this chapter.** The chemistry share of the CBT-2 "Physics & Chemistry" section is pitched at NCERT Class 10–12 level, dominated by Class 10 topics (matter, acids-bases-salts, metals & non-metals, carbon) with a few Class 11 fundamentals (mole concept, atomic structure, periodic trends, bonding). Every fact here is aligned with NCERT Science (Class 9 & 10) and NCERT Chemistry (Class 11 & 12). Study the theory, the 10 worked examples, the PYQ Corner (real 2025 CBT-2 questions), then the 25 MCQs and the revision tables.

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1. Matter & Its States

1.1 States of Matter (NCERT Class 9, Ch. Matter in Our Surroundings)

Property	Solid	Liquid	Gas
Shape	Fixed	Takes container's	Takes container's
Volume	Fixed	Fixed	Fills container
Compressibility	Negligible	Very low	High
Intermolecular force	Strongest	Moderate	Weakest
Interparticle space	Least	Moderate	Largest

Two additional states: **plasma** (superheated ionised gas — the Sun, neon signs) and **Bose–Einstein condensate** (near absolute zero).

1.2 Changes of State

- **Melting/fusion** (solid → liquid at melting point), **boiling/vaporisation**, **condensation**, **freezing**.
- **Sublimation:** solid → gas directly (camphor, naphthalene, dry ice, iodine, ammonium chloride).
- **Deposition:** gas → solid directly.
- Latent heat is absorbed during melting and boiling **at constant temperature**.

Evaporation (surface phenomenon, all temperatures) causes cooling and increases with surface area, temperature, and wind, and decreases with humidity.

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2. Elements, Compounds & Mixtures

2.1 Classification of Matter (NCERT Class 9, Ch. Is Matter Around Us Pure?)

- **Element:** made of one kind of atom (e.g., Fe, O_2, S_8). ~118 known; metals, non-metals, metalloids.
- **Compound:** two or more elements chemically combined in a **fixed ratio**, with properties different from constituents (e.g., H_2O, NaCl, CO_2). Components separated only by chemical means.
- **Mixture:** two or more substances physically mixed in **any ratio**, retaining their own properties; separated by physical methods.

2.2 Mixtures vs Compounds (a favourite exam contrast)

Feature	Mixture	Compound
Combination	Physical	Chemical
Ratio of components	Variable	Fixed & definite
Properties	Those of components	New, different from components
Energy change on forming	Usually none	Heat/light usually evolved/absorbed
Separation	Physical methods	Chemical methods only
Example	Air, brass, salt solution	Water, common salt, sugar

2.3 Homogeneous vs Heterogeneous Mixtures

- **Homogeneous:** uniform composition throughout, single phase — solutions (salt in water, air, **alloys such as brass, bronze, steel, and stainless steel**).
- **Heterogeneous:** non-uniform, distinct components/phases — sand + iron filings, oil + water, muddy water.

> **Key exam fact: Alloys are homogeneous mixtures** of two or more metals (or a metal and a non-metal), e.g., brass (Cu + Zn), bronze (Cu + Sn), steel (Fe + C). They are prepared by melting the components together and mixing at the atomic level, so their composition is uniform.

2.4 Solutions — Terminology

- **Solute** (smaller amount) dissolves in **solvent** (larger amount). Water is the "universal solvent".
- **Solubility** = maximum solute that dissolves in 100 g solvent at a given temperature; a **saturated** solution can dissolve no more at that temperature.
- **Tyndall effect** distinguishes colloids/suspensions (scatter light) from true solutions (don't).

Type	Particle size	Example	Filterable?	Tyndall?
True solution	< 1 nm	Salt water	No	No
Colloid	1–1000 nm	Milk, fog	No (needs ultra-)	Yes
Suspension	> 1000 nm	Chalk in water	Yes	Yes

2.5 Separation Techniques

Method	Separates	Basis
Filtration	Insoluble solid from liquid	Particle size
Evaporation	Soluble solid from liquid	Volatility of solvent
Distillation	Miscible liquids with different b.p.	Boiling point
Fractional distillation	Two miscible liquids with close b.p. (petroleum)	Boiling point + fractionating column
Separating funnel	Immiscible liquids (oil + water)	Density/immiscibility
Sublimation	Sublimable solid from a mixture	Sublimation
Centrifugation	Suspended particles (cream from milk)	Density under spin
Chromatography	Coloured components of a mixture (dyes, pigments)	Different adsorption/solubility on a medium
Crystallisation	Pure solid from solution	Differential crystal formation

Chromatography (Greek *chroma* = colour): separates dissolved constituents that move at different rates on paper/adsorbent — used to separate ink dyes and plant pigments.

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3. Mole Concept & Stoichiometry

3.1 Fundamental Quantities (NCERT Class 9 & 11)

- **Atomic mass unit (u):** 1 u = 1/12 the mass of a carbon-12 atom.
- **Mole:** amount containing **Avogadro's number N_A = 6.022 × 10^{23}** particles.
- **Molar mass:** mass of 1 mole in grams = numerically equal to atomic/molecular mass in u.
- Number of moles **n = given mass ÷ molar mass = number of particles ÷ N_A = volume of gas (STP) ÷ 22.4 L**.

3.2 Molecular Mass — Worked Method

Add the atomic masses (from the periodic table) of all atoms in the formula.

- H_2O = 2(1) + 16 = **18 u**
- CO_2 = 12 + 2(16) = **44 u**
- H_2SO_4 = 2 + 32 + 4(16) = **98 u**
- **S_8 = 8 × 32 = 256 u** (sulphur exists as an S_8 ring; this appears in CBT-2)

3.3 Concentration Terms

Term	Formula
Mass percent (w/w)	(mass of solute ÷ mass of solution) × 100
Volume percent (v/v)	(volume of solute ÷ volume of solution) × 100
Molarity (M)	moles of solute ÷ volume of solution in litres
Molality (m)	moles of solute ÷ mass of solvent in kg
Mole fraction	moles of component ÷ total moles

Note: *mass of solution = mass of solute + mass of solvent.* This is the exact point CBT-2 tests.

3.4 Gas Laws (Class 11 basics)

- **Boyle's law:** at constant T, PV = constant ($P \propto 1/V$).
- **Charles's law:** at constant P, V/T = constant ($V \propto T$ in kelvin).
- **Gay-Lussac's law:** at constant V, P/T = constant.
- **Avogadro's law:** equal volumes of all gases at the same T and P contain equal numbers of molecules. At **STP (0 °C, 1 atm), 1 mole of any gas = 22.4 L**.
- **Ideal gas equation: PV = nRT** ($R = 8.314\ J \cdot mol^{-1} \cdot K^{-1}$).

Worked Example 1. How many moles and how many molecules are there in 36 g of water?

Molar mass of H_2O = 18 g/mol. n = 36/18 = **2 mol**.

Molecules = $2 \times 6.022 \times 10^{23}$ = **1.2044×10^{24}**.

Worked Example 2. Find the mass percent of a solution made by dissolving 40 g of NaCl in 200 g of water.

Mass of solution = 40 + 200 = 240 g.

Mass % = (40/240) × 100 = **16.67 %**. *(This is a verbatim 2025 CBT-2 question — note that the denominator is the mass of the solution, not just the water.)*

Worked Example 3. Calculate the molar mass of glucose ($C_6H_{12}O_6$).

= 6(12) + 12(1) + 6(16) = 72 + 12 + 96 = **180 g/mol**.

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4. Atomic Structure

4.1 Subatomic Particles (NCERT Class 9, Ch. Structure of the Atom)

Particle	Charge	Relative mass (u)	Discoverer
Electron	−1	1/1836 (≈ 0)	J. J. Thomson
Proton	+1	1	Goldstein / Rutherford
Neutron	0	1	James Chadwick

- **Atomic number (Z)** = number of protons = number of electrons in a neutral atom (defines the element).
- **Mass number (A)** = protons + neutrons; so **neutrons = A − Z**.
- **Isotopes:** same Z, different A (^{1}H, ^{2}H, ^{3}H; ^{12}C, ^{14}C).
- **Isobars:** same A, different Z (^{40}Ar, ^{40}Ca). **Isotones:** same number of neutrons.

4.2 Bohr Model & Electron Shells

- Electrons revolve in fixed **shells (K, L, M, N…)** of definite energy; no energy radiated while in a shell.
- Maximum electrons in a shell = **$2n^2$** (K=2, L=8, M=18, N=32); outermost shell ≤ 8 (octet).
- **Valence electrons** in the outermost shell decide chemical behaviour and valency.

4.3 Quantum Numbers (Class 11)

Quantum number	Symbol	Describes
Principal	n	Shell/energy level & size (1, 2, 3…)
Azimuthal	l	Subshell/shape (0=s, 1=p, 2=d, 3=f)
Magnetic	m_l	Orbital orientation (−l … +l)
Spin	m_s	Electron spin (+½ or −½)

Subshell capacities: **s = 2, p = 6, d = 10, f = 14** electrons.

4.4 Rules for Electronic Configuration

- **Aufbau principle:** electrons fill orbitals in order of increasing energy (1s, 2s, 2p, 3s, 3p, 4s, 3d…).
- **Pauli exclusion principle:** no two electrons in an atom have all four quantum numbers identical (an orbital holds at most 2 electrons of opposite spin).
- **Hund's rule:** orbitals of a subshell are singly filled (parallel spins) before pairing.

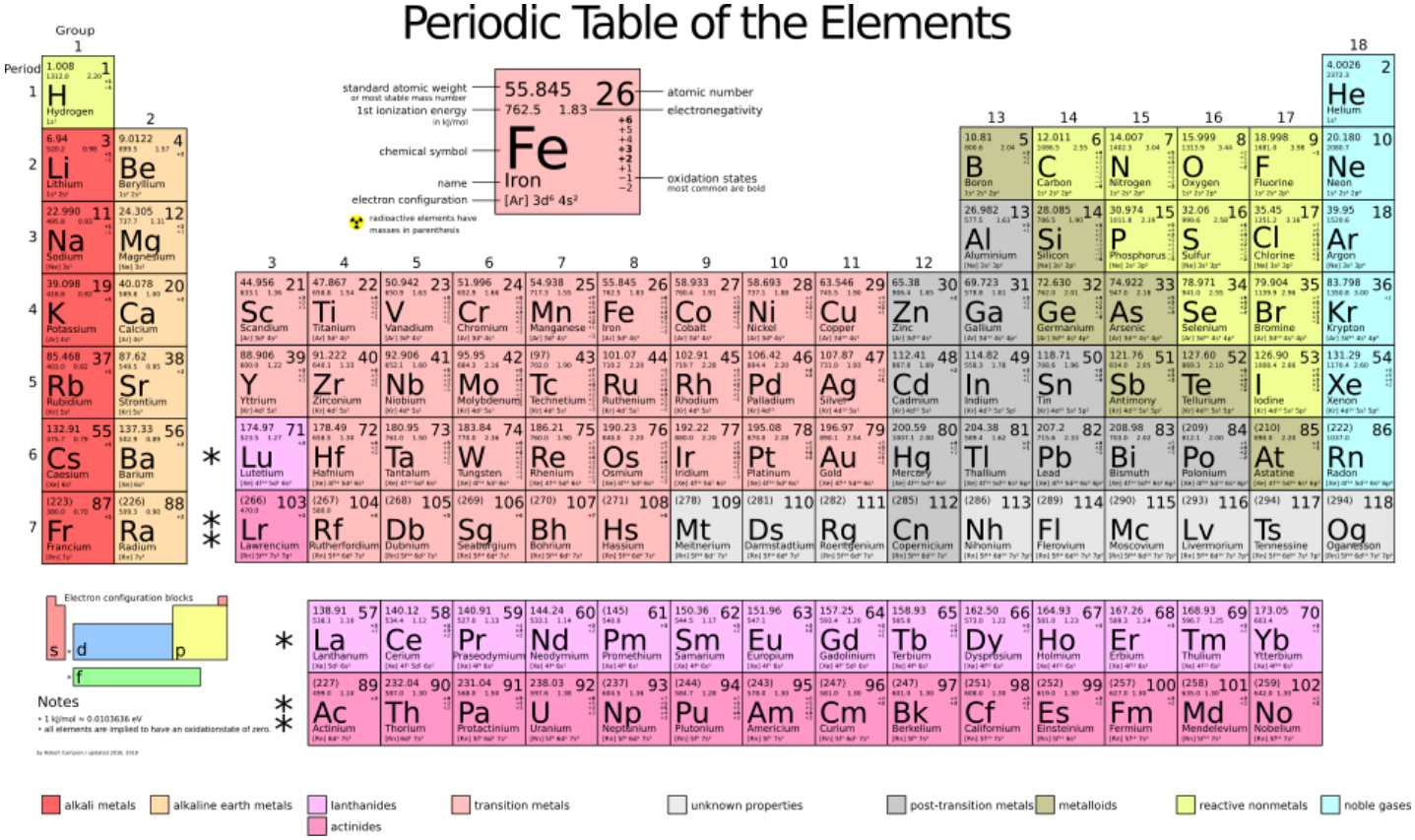
Example configurations: Na (11) = 2, 8, 1 → $1s^2 2s^2 2p^6 3s^1$; Cl (17) = 2, 8, 7; **O (8) = 2, 6 → $1s^2 2s^2 2p^4$** (element = oxygen).

> **Worked Example 4.** An element has atomic number 8. Identify it and give its electronic configuration and valency.

> Z = 8 ⇒ **oxygen**. Configuration 2, 6 ($1s^2 2s^2 2p^4$). It needs 2 electrons to complete the octet ⇒ **valency 2**. *(Direct 2025 CBT-2 question.)*

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5. Periodic Classification



Modern periodic table.

5.1 Development (NCERT Class 10, Ch. Periodic Classification)

- **Döbereiner's triads** → **Newlands' law of octaves** → **Mendeleev's periodic table** (properties = periodic function of atomic **mass**; predicted eka-boron, eka-silicon).
- **Modern periodic law (Moseley):** properties are a periodic function of **atomic number (Z)**.
- Modern table: **18 groups, 7 periods**; blocks s, p, d (transition), f (lanthanides & actinides — inner transition).

5.2 Periodic Trends

Property	Along a period (left → right)	Down a group
Atomic size/radius	Decreases	Increases
Metallic character	Decreases	Increases
Non-metallic character	Increases	Decreases
Ionisation energy	Increases	Decreases
Electron affinity	Increases (generally)	Decreases
Electronegativity	Increases	Decreases
Valence electrons	Increase 1→8	Same in a group

- Most metallic element region: bottom-left; most non-metallic (excluding noble gases): top-right (**fluorine** is the most electronegative element).
- Group 1 = alkali metals; Group 2 = alkaline earth metals; Group 17 = halogens; Group 18 = noble/inert gases (stable octet).

6. Chemical Bonding

6.1 Why Atoms Bond

To attain a **stable noble-gas (octet/duplet)** configuration, atoms lose, gain, or share electrons.

6.2 Ionic vs Covalent Bonds (NCERT Class 10, Ch. Metals & Non-Metals)

Property	Ionic (electrovalent)	Covalent
Formation	Transfer of electrons (metal → non-metal)	Sharing of electrons (non-metal + non-metal)
Example	NaCl, MgO, $CaCl_2$	H_2, O_2, CH_4, H_2O, CO_2
Physical state	Solids (crystalline)	Gases, liquids, or soft solids
Melting/boiling point	**High** (strong electrostatic lattice)	**Low** — held by **weak intermolecular forces**
Solubility	Soluble in water (polar)	Generally soluble in organic solvents
Electrical conductivity	Conduct when molten/aqueous (free ions)	Non-conductors (no free ions/electrons)

> **Key exam fact: Covalent compounds have low melting and boiling points** because, although the intramolecular covalent bonds are strong, the **intermolecular forces between molecules are weak**, so little energy is needed to separate the molecules.

6.3 Hybridization (Class 11)

Molecule	Central atom hybridization	Shape	Bond angle
$BeCl_2$	sp	Linear	180°
BF_3	sp^2	Trigonal planar	120°
CH_4	sp^3	Tetrahedral	109.5°
NH_3	sp^3 (1 lone pair)	Trigonal pyramidal	~107°
H_2O	sp^3 (2 lone pairs)	Bent/angular	~104.5°

Lone pairs compress the bond angle (CH_4 109.5° > NH_3 107° > H_2O 104.5°).

6.4 Hydrogen Bonding

A weak electrostatic attraction between an H atom bonded to a highly electronegative atom (F, O, N) and another electronegative atom. It explains:

- The **high boiling point of water** (vs H_2S) and its liquid state at room temperature.
- Ice being **less dense than water** (open hydrogen-bonded lattice → ice floats).
- The structure of DNA and proteins.

7. Chemical Reactions

7.1 Types (NCERT Class 10, Ch. Chemical Reactions & Equations)

Type	General form	Example
Combination	A + B → AB	$CaO + H_2O \rightarrow Ca(OH)_2$
Decomposition	AB → A + B	$2H_2O \rightarrow 2H_2 + O_2$ (electrolysis); thermal/photo
Displacement	A + BC → AC + B	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Double displacement	AB + CD → AD + CB	$AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$ (precipitation)
Neutralisation	acid + base → salt + water	$HCl + NaOH \rightarrow NaCl + H_2O$

- **Exothermic** reactions release heat (respiration, combustion, quicklime + water); **endothermic** absorb heat (photosynthesis, thermal decomposition).
- **Balancing:** the number of atoms of each element must be equal on both sides (law of conservation of mass).

7.2 Redox Reactions & Oxidation Number

- **Oxidation:** loss of electrons / gain of oxygen / loss of hydrogen; **reduction:** gain of electrons / loss of oxygen / gain of hydrogen. Remember **OIL RIG** (Oxidation Is Loss, Reduction Is Gain of electrons).
- **Oxidising agent** gets reduced; **reducing agent** gets oxidised.
- **Rules for oxidation number:** free element = 0; O = −2 (−1 in peroxides); H = +1 (−1 in metal hydrides); the sum equals the overall charge.
- **Corrosion** (rusting of iron) and **rancidity** (oxidation of fats/oils) are everyday redox processes.

Worked Example 5. Find the oxidation number of sulphur in H_2SO_4.

Let it be x. $2(+1) + x + 4(-2) = 0 \Rightarrow 2 + x - 8 = 0 \Rightarrow$ **x = +6**.

Worked Example 6. Balance: $Fe + O_2 \rightarrow Fe_2O_3$.

$4Fe + 3O_2 \rightarrow 2Fe_2O_3$ (8 Fe? no — 4 Fe and 6 O on each side). Check: LHS 4 Fe, 6 O; RHS 2×2 = 4 Fe, 2×3 = 6 O. ✓

8. Acids, Bases & Salts

8.1 Definitions (NCERT Class 10, Ch. Acids, Bases & Salts)

- **Acids** give H^+ (H_3O^+) in water — sour, turn blue litmus red (HCl, H_2SO_4, HNO_3, citric acid).
- **Bases** give OH^- in water — bitter, soapy, turn red litmus blue; a water-soluble base is an **alkali** (NaOH, KOH, $Ca(OH)_2$).
- Acid + metal → salt + **hydrogen gas** (burns with a "pop"). Acid + metal carbonate/bicarbonate → salt + water + **CO_2** (turns lime water milky).

8.2 The pH Scale

$pH = -\log[H^+]$, ranging **0 to 14** at 25 °C.

pH	Nature
< 7	Acidic (lower pH → stronger acid)
= 7	Neutral (pure water)
> 7	Basic/alkaline

Daily-life values: gastric juice ~1.2, lemon juice ~2.2, vinegar ~3, black coffee ~5, milk ~6.5, pure water 7, blood ~7.4, baking-soda solution ~9, milk of magnesia ~10.

> **Key exam fact:** Adding **lemon juice (a citric-acid solution) to water lowers its pH below 7** (makes it acidic), because the H^+ concentration rises.

Importance of pH: tooth decay below pH 5.5 (bacteria act on sugar → acids); antacids (milk of magnesia) neutralise excess stomach acid; soil pH affects crops; bee-sting (acidic) soothed by baking soda, wasp-sting (basic) by vinegar.

8.3 Indicators

Indicator	In acid	In base
Litmus	Red	Blue
Methyl orange	Red/pink	Yellow
Phenolphthalein	Colourless	Pink
Turmeric	Yellow	Red/brown

Olfactory indicators: onion, vanilla, clove (change smell).

8.4 Neutralisation

Acid + Base → Salt + Water (a specific double-displacement reaction), e.g., HCl + NaOH → NaCl + H_2O. Strong acid + strong base gives a **neutral** salt; strong acid + weak base → acidic salt; weak acid + strong base → basic salt.

8.5 Important Salts — Formula, Preparation & Uses

Common name	Chemical name	Formula	Key use
Common salt	Sodium chloride	NaCl	Food; raw material for NaOH, $NaHCO_3$
Caustic soda	Sodium hydroxide	NaOH	Soap, paper (from chlor-alkali process)
Baking soda	Sodium hydrogen carbonate	$NaHCO_3$	Antacid, baking, fire extinguishers
Washing soda	Sodium carbonate decahydrate	$Na_2CO_3 \cdot 10H_2O$	Cleaning, softening hard water, glass
Bleaching powder	Calcium oxychloride	$CaOCl_2$	Bleaching, disinfecting water
Plaster of Paris	Calcium sulphate hemihydrate	$CaSO_4 \cdot \frac{1}{2}H_2O$	Plaster casts, ceilings, moulds
Gypsum	Calcium sulphate dihydrate	$CaSO_4 \cdot 2H_2O$	Raw material for PoP, cement
Blue vitriol	Copper sulphate pentahydrate	$CuSO_4 \cdot 5H_2O$	Fungicide, electroplating
Washing soda / lime	Slaked lime	$Ca(OH)_2$	Whitewashing, mortar
Quicklime	Calcium oxide	CaO	Cement, steel, whitewash

- **Water of crystallisation:** fixed water molecules in a crystal ($CuSO_4 \cdot 5H_2O$ is blue; on heating it turns white anhydrous).
- **Chlor-alkali process:** electrolysis of brine → NaOH + Cl_2 + H_2.
- On heating gypsum to ~373 K → **Plaster of Paris**; PoP + water → gypsum (sets hard).

> **Worked Example 7.** Why does milk of magnesia relieve acidity?

> It is a mild base, $Mg(OH)_2$. It **neutralises** excess HCl in the stomach: $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$, raising the pH toward normal.

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9. Carbon & Its Compounds

9.1 Bonding & Allotropes (NCERT Class 10, Ch. Carbon and Its Compounds)

- Carbon (Z = 6, config 2,4) forms **four covalent bonds** (tetravalency) and shows **catenation** (self-linking into long chains, branches, and rings) — the reason for millions of carbon compounds.
- **Allotropes: diamond** (each C bonded to 4 others, hardest natural substance, non-conductor), **graphite** (layered, soft, slippery, conducts electricity — used as electrodes/lubricant), **fullerene/buckminsterfullerene C_{60}** (spherical), graphene, carbon nanotubes.

9.2 Hydrocarbons

Series	General formula	Bond	Example
Alkanes (saturated)	C_nH_{2n+2}	Single	CH_4 methane, C_2H_6 ethane
Alkenes (unsaturated)	C_nH_{2n}	One double	C_2H_4 ethene
Alkynes (unsaturated)	C_nH_{2n-2}	One triple	C_2H_2 ethyne (acetylene)

- **Saturated** hydrocarbons burn with a clean blue flame; **unsaturated** ones give a sooty yellow flame and undergo **addition** reactions (hydrogenation of oils → vanaspati using Ni catalyst).
- **Homologous series:** compounds with the same general formula and functional group, differing by **$–CH_2–$ (14 u)**; they show a gradation in physical properties and similar chemical properties.

9.3 Functional Groups

Group	Formula	Class	Example
Halo	–X (Cl, Br)	Haloalkane	Chloroethane
Hydroxyl	–OH	Alcohol	Ethanol C_2H_5OH
Aldehyde	–CHO	Aldehyde	Ethanal
Ketone	>C=O	Ketone	Propanone (acetone)
Carboxyl	–COOH	Carboxylic acid	Ethanoic acid CH_3COOH (acetic acid; vinegar)

9.4 Isomerism

Compounds with the same molecular formula but different structures (hence different properties), e.g., **butane and isobutane (C_4H_{10})**. Isomerism starts from C_4 in alkanes.

9.5 Important Compounds

- **Ethanol (C_2H_5OH):** alcohol in beverages; solvent; denatured with methanol for industrial use.
- **Ethanoic acid (CH_3COOH):** 5–8% solution = **vinegar**; freezes to "glacial" acetic acid.
- Soaps (sodium salts of fatty acids) clean by forming **micelles**; detergents work even in hard water.

9.6 Polymers, Carbohydrates, Proteins & Fats (Class 12 basics)

Polymer	Type	Use
Polythene	Addition (from ethene)	Bags, packaging
PVC (polyvinyl chloride)	Addition	Pipes, insulation, raincoats
Nylon	Condensation	Ropes, fabrics, parachutes
Terylene/PET	Condensation (polyester)	Fabrics, bottles
Bakelite	Thermosetting	Switches, handles
Natural rubber	Polymer of isoprene; **vulcanised** with sulphur for strength	Tyres

- **Carbohydrates** (C, H, O): monosaccharides (glucose, fructose), disaccharides (sucrose, lactose), polysaccharides (starch, cellulose, glycogen); primary energy source.
- **Proteins:** polymers of **amino acids** (peptide bonds); enzymes, muscle, structural roles.
- **Fats/lipids:** esters of glycerol and fatty acids; energy store; saturated (animal) vs unsaturated (vegetable) fats.

> **Worked Example 8.** Write the molecular formula and name of the first three members of the alkane series.

> Methane **CH_4**, Ethane **C_2H_6**, Propane **C_3H_8** (general formula C_nH_{2n+2}).

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10. Metallurgy

10.1 Physical Properties of Metals (NCERT Class 10, Ch. Metals & Non-Metals)

Metals are lustrous, **malleable** (beaten into sheets), **ductile** (drawn into wires), sonorous, good conductors of heat and electricity, and generally solid with high melting points.

- Most ductile/malleable: **gold and silver**. Best conductor: **silver**, then copper.
- Exceptions: mercury is liquid; sodium/potassium are soft (cut with a knife); non-metals are generally brittle, dull, poor conductors (except **graphite**, which conducts; **diamond**, hardest).

10.2 Reactivity (Activity) Series

K > Na > Ca > Mg > Al > Zn > Fe > Pb > (H) > Cu > Hg > Ag > Au (decreasing reactivity).

- A more reactive metal displaces a less reactive one from its salt solution.
- Metals above hydrogen displace H_2 from dilute acids; those below (Cu, Ag, Au) do not.

10.3 Extraction Flow (depends on reactivity)

Reactivity	Occurrence	Extraction method
High (K, Na, Ca, Mg, Al)	Combined	**Electrolysis** of molten ore
Medium (Zn, Fe, Pb)	Oxides/carbonates/sulphides	Roasting/calcination → **reduction** (with C/CO or thermite)
Low (Cu, Ag, Au, Hg)	Often **free (native)**	Simple heating/roasting of ore

Key steps: **concentration of ore** → **roasting** (sulphide, in air) / **calcination** (carbonate, limited air) → **reduction to metal** → **refining (electrolytic)**. **Thermite reaction:** $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + heat (used to weld rail tracks).

10.4 Corrosion & Prevention

- **Rusting** of iron (hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$) needs **both air (oxygen) and moisture**. Copper forms a green carbonate coating; silver blackens (sulphide).
- **Prevention:** painting, oiling/greasing, **galvanisation** (zinc coating), tinning, **electroplating**, alloying, and cathodic (sacrificial) protection.

10.5 Alloys — Composition & Uses

Alloy	Composition	Use
Brass	Cu + Zn	Utensils, fittings
Bronze	Cu + Sn	Statues, medals, bells
Steel	Fe + C (up to ~2%)	Construction
Stainless steel	Fe + Cr + Ni	Cutlery, utensils (rust-resistant)
Solder	Pb + Sn	Joining/soldering wires (low m.p.)
Duralumin	Al + Cu + Mg + Mn	Aircraft bodies (light & strong)
Amalgam	Any metal + Hg	Dental fillings
German silver	Cu + Zn + Ni	Ornaments (no silver in it)

Alloys are **homogeneous mixtures**; alloying improves hardness, corrosion resistance, and lowers melting point (solder).

> **Worked Example 9.** Why is zinc coated on iron objects in galvanisation?

> Zinc is **more reactive** than iron; it corrodes preferentially (sacrificial protection) and its coating physically blocks air and moisture from the iron, preventing rust even if the layer is scratched.

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11. Environmental Chemistry

11.1 Air Pollution & Acid Rain

- Oxides of sulphur and nitrogen (**SO_2, NO_2**) from fossil-fuel burning dissolve in rain to form **sulphuric and nitric acids** → **acid rain** (pH < 5.6): damages buildings (marble — "stone cancer"), soil, and aquatic life.

11.2 Ozone Depletion

- The stratospheric **ozone (O_3)** layer absorbs harmful **UV** radiation. **CFCs (chlorofluorocarbons)** release chlorine that destroys ozone, creating the "ozone hole" (over Antarctica). The **Montreal Protocol (1987)** phases out CFCs.

11.3 Greenhouse Effect & Global Warming

- Greenhouse gases (**CO_2, CH_4, N_2O, water vapour, CFCs**) trap infrared radiation and warm the Earth. Excess emissions → **global warming**, climate change, rising sea levels. Governed globally by frameworks like the Paris Agreement.

11.4 Green Chemistry

Designing chemical processes that reduce or eliminate hazardous substances — atom economy, safer solvents, renewable feedstocks, less waste (the basis of sustainable "green" manufacturing).

> **Worked Example 10.** Why is acid rain harmful to the Taj Mahal?

> The Taj is made of **marble ($CaCO_3$)**. Acids in rain react: $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$, corroding and yellowing the marble ("marble/stone cancer").

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12. PYQ Corner — RRB JE CBT-2 2025 (Chemistry Questions Solved)

PYQ 1. Alloys are examples of which type of mixture?

Homogeneous mixtures — metals melted together mix uniformly at the atomic level (e.g., brass = Cu + Zn), giving a single-phase, uniform composition (§2.3).

PYQ 2. 40 g of NaCl is dissolved in 200 g of water. Find the mass percent of the solution.

Mass of solution = 40 + 200 = 240 g. Mass % = (40/240) × 100 = **16.67 %** (§3.3, W.E. 2).

PYQ 3. An element has atomic number 8. Name it.

Z = 8 ⇒ **oxygen** (configuration 2, 6; valency 2) (§4.4, W.E. 4).

PYQ 4. Why do covalent compounds have low melting and boiling points?

Because the **intermolecular forces between covalent molecules are weak**, so little energy is needed to separate them — even though the intramolecular covalent bonds themselves are strong (§6.2).

PYQ 5. Calculate the molecular mass of S_8.

Atomic mass of S = 32 u. S_8 = 8 × 32 = **256 u** (§3.2).

PYQ 6. What happens to the pH of water when a few drops of lemon juice are added?

Lemon juice contains citric acid; it releases H^+ ions, so the **pH decreases below 7** (the water becomes acidic) (§8.2).

PYQ 7. The property by which a metal can be drawn into thin wires is called:

Ductility (drawing into wires); malleability = beaten into sheets (§10.1). *(Shared with the physics/materials PYQ set.)*

PYQ 8. Differentiate a mixture from a compound (one key point).

A **compound** has a fixed composition with new properties and is separable only chemically; a **mixture** has a variable composition, retains component properties, and is separated by physical methods (§2.2).

13. Practice MCQs (25)

Q1. Which of the following is a homogeneous mixture?

(a) sand + water (b) oil + water (c) brass (d) iron filings + sulphur

Q2. The process of a solid changing directly into vapour is:

(a) condensation (b) sublimation (c) evaporation (d) fusion

Q3. Which technique separates the coloured components of a dye?

(a) filtration (b) distillation (c) chromatography (d) centrifugation

Q4. The number of molecules in 1 mole of any substance is:

(a) 6.022×10^{22} (b) 6.022×10^{23} (c) 3.011×10^{23} (d) 1.6×10^{19}

Q5. The molar mass of CO_2 is:

(a) 22 g (b) 28 g (c) 44 g (d) 48 g

Q6. 20 g of sugar is dissolved in 80 g of water. The mass percent of sugar is:

(a) 16 % (b) 20 % (c) 25 % (d) 80 %

Q7. The volume occupied by 1 mole of any gas at STP is:

(a) 11.2 L (b) 22.4 L (c) 24 L (d) 6.022 L

Q8. The neutron was discovered by:

(a) Rutherford (b) J. J. Thomson (c) Chadwick (d) Bohr

Q9. The maximum number of electrons in the M shell is:

(a) 2 (b) 8 (c) 18 (d) 32

Q10. An atom with atomic number 8 has valency:

(a) 1 (b) 2 (c) 6 (d) 8

Q11. The modern periodic law is based on:

(a) atomic mass (b) atomic number (c) valency (d) density

Q12. The most electronegative element is:

(a) oxygen (b) chlorine (c) fluorine (d) nitrogen

Q13. Down a group in the periodic table, atomic size:

(a) decreases (b) increases (c) stays the same (d) first increases then decreases

Q14. Covalent compounds generally have:

(a) high melting points (b) low melting points (c) high conductivity in solid state (d) ionic lattices

Q15. The shape of a methane (CH_4) molecule is:

(a) linear (b) trigonal planar (c) tetrahedral (d) bent

Q16. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ is a:

(a) combination reaction (b) decomposition (c) displacement (d) double displacement

Q17. The oxidation number of sulphur in H_2SO_4 is:

(a) +2 (b) +4 (c) +6 (d) −2

Q18. A solution turns phenolphthalein pink. It is:

(a) acidic (b) basic (c) neutral (d) a salt of a strong acid

Q19. Adding lemon juice to water makes its pH:

(a) increase above 7 (b) decrease below 7 (c) stay exactly 7 (d) reach 14

Q20. The chemical name of baking soda is:

(a) sodium carbonate (b) sodium hydroxide (c) sodium hydrogen carbonate (d) calcium carbonate

Q21. Plaster of Paris is:

(a) $CaSO_4 \cdot 2H_2O$ (b) $CaSO_4 \cdot \frac{1}{2}H_2O$ (c) $CaCO_3$ (d) CaO

Q22. The general formula of alkanes is:

(a) C_nH_{2n} (b) C_nH_{2n-2} (c) C_nH_{2n+2} (d) C_nH_n

Q23. The property of carbon to form long chains by self-linking is called:

(a) tetravalency (b) catenation (c) isomerism (d) hybridisation

Q24. Coating iron with zinc to prevent rust is called:

(a) tinning (b) galvanisation (c) alloying (d) annealing

Q25. Acid rain is mainly caused by oxides of:

(a) carbon and hydrogen (b) sulphur and nitrogen (c) sodium and calcium (d) oxygen and helium

Answer Key with One-Line Reasons

Q	Ans	Reason
1	c	Brass (Cu+Zn) is a uniform, single-phase alloy
2	b	Solid → vapour directly = sublimation
3	c	Chromatography separates by differential adsorption
4	b	Avogadro's number = 6.022×10^{23}
5	c	$12 + 2(16) = 44$ g/mol
6	b	$20/(20+80) \times 100 = 20$ %
7	b	Molar volume at STP = 22.4 L
8	c	James Chadwick discovered the neutron (1932)
9	c	$2n^2 = 2(3^2) = 18$
10	b	Oxygen (2,6) needs 2 electrons → valency 2
11	b	Properties are periodic functions of atomic number
12	c	Fluorine is the most electronegative element
13	b	New shells added → size increases down a group
14	b	Weak intermolecular forces → low m.p./b.p.
15	c	sp^3 hybridisation → tetrahedral
16	c	Fe displaces less reactive Cu from its salt
17	c	$2(+1) + x + 4(-2) = 0 \rightarrow x = +6$
18	b	Phenolphthalein is pink only in bases
19	b	Citric acid raises $[H^+]$, lowering pH
20	c	$NaHCO_3$ = sodium hydrogen carbonate
21	b	PoP = $CaSO_4 \cdot \frac{1}{2}H_2O$
22	c	Saturated alkanes: C_nH_{2n+2}
23	b	Catenation = self-linking of carbon
24	b	Zinc coating = galvanisation
25	b	SO_2 and NO_2 form H_2SO_4/HNO_3 in rain

14. Rapid-Revision Tables

Mole & Concentration

- n = mass/molar mass = particles/N_A = volume(STP)/22.4 L; N_A = 6.022×10^{23}
- **Mass % = (mass solute / mass solution) × 100**; mass solution = solute + solvent
- Molarity = mol/L solution; Molality = mol/kg solvent; PV = nRT; molar volume 22.4 L at STP

- Molecular masses to memorise: H_2O 18, CO_2 44, NaCl 58.5, H_2SO_4 98, glucose 180, **S_8 256**

Atomic Structure

- neutrons = A − Z; shell capacity $2n^2$; subshells s2 p6 d10 f14
- O (8) = 2,6; Na (11) = 2,8,1; Cl (17) = 2,8,7
- Aufbau (energy order) · Pauli (no identical 4 quantum numbers) · Hund (singly fill first)

Bonding

- Ionic → transfer, high m.p., conduct when molten/aqueous
- **Covalent → sharing, low m.p./b.p. (weak intermolecular forces), non-conducting**
- Shapes: $BeCl_2$ linear, BF_3 trigonal planar, CH_4 tetrahedral, NH_3 pyramidal, H_2O bent

Acids, Bases & Salts

- pH 0–14; **< 7 acidic, 7 neutral, > 7 basic**; lemon juice lowers pH
- Litmus: acid→red, base→blue; phenolphthalein pink in base
- Baking soda $NaHCO_3$ · Washing soda $Na_2CO_3{\cdot}10H_2O$ · Bleaching powder $CaOCl_2$ · PoP $CaSO_4{\cdot}½H_2O$ · Caustic soda NaOH

Carbon

- Tetravalency + catenation; allotropes diamond/graphite/fullerene
- Alkane C_nH_{2n+2}, alkene C_nH_{2n}, alkyne C_nH_{2n-2}; homologous difference $–CH_2–$ (14 u)
- Functional groups: –OH alcohol, –CHO aldehyde, –COOH acid, >C=O ketone

Metallurgy & Environment

- Activity series: K Na Ca Mg Al Zn Fe Pb (H) Cu Hg Ag Au
- High reactivity → electrolysis; medium → reduction; low → native/heating
- Rusting needs air + moisture; prevent by galvanising, painting, alloying
- Alloys are **homogeneous mixtures**: brass Cu+Zn, bronze Cu+Sn, steel Fe+C, stainless steel Fe+Cr+Ni
- Acid rain: SO_2/NO_2 · Ozone hole: CFCs · Greenhouse: CO_2/CH_4

End of GA-02. See GA-01 for the Physics companion chapter.